

KARASUK, Russian Federation

WMO: 298140

Lat: 53.7345N

Lon: 78.0137E

Elev: 377

StdP: 14.50

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.5	-25.0	-36.1	0.7	-30.5	-30.6	1.0	-24.7	27.1	18.3	23.2	19.9	4.3	240	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	87.7	65.1	84.9	64.6	81.8	63.5	68.7	80.6	67.2	78.6	65.8	77.2	8.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	93.6	73.1	63.2	87.9	72.0	61.6	83.0	71.0	33.3	80.6	31.9	78.4	30.9	77.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.8	17.3	-34.8	93.7	7.1	2.9	-40.0	95.8	-44.1	97.6	-48.1	99.2	-53.3	101.3
			-34.6	71.0	6.9	2.0	-39.5	72.5	-43.5	73.6	-47.4	74.8	-52.4	76.2
				DB										
				WB										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	-0.4	4.0	19.3	41.5	56.1	66.2	68.7	64.4	53.7	39.7	21.2	7.0
	DBStd	26.76	14.47	13.01	11.64	10.45	9.64	7.39	5.98	6.76	8.40	8.97	13.45	13.89
	HDD50	6742	1563	1289	953	294	52	2	0	1	55	337	864	1332
	HDD65	10609	2028	1709	1418	705	302	75	33	95	349	784	1314	1797
	CDD50	1983	0	0	0	40	242	488	581	447	167	18	0	0
	CDD65	375	0	0	0	0	27	112	148	76	12	0	0	0
	CDH74	3543	0	0	0	15	351	1021	1245	728	181	2	0	0
	CDH80	1026	0	0	0	1	88	311	368	206	52	0	0	0
Wind	WSAvg	7.7	7.8	8.1	8.5	8.5	8.3	7.1	6.3	6.4	6.9	7.6	8.3	8.7
Precipitation	PrecAvg	12.5	0.5	0.4	0.6	0.6	0.9	1.7	2.6	1.6	0.9	1.1	0.8	0.7
	PrecMax	20.0	1.1	1.1	1.7	1.8	2.5	5.0	6.1	4.1	2.0	2.5	1.8	1.6
	PrecMin	7.7	0.0	0.0	0.1	0.0	0.0	0.2	1.0	0.3	0.2	0.1	0.5	0.3
	PrecStd	2.8	0.3	0.3	0.4	0.4	0.6	1.1	1.5	0.8	0.5	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.9	33.6	45.9	75.8	87.9	90.8	91.5	90.4	86.7	69.5	48.9	35.0
		MCWB	32.7	31.6	39.4	56.8	61.6	65.3	67.3	66.0	62.1	52.5	43.3	32.8
	2%	DB	29.3	30.8	39.6	69.2	82.1	87.0	87.0	85.5	78.9	63.3	44.0	32.7
		MCWB	27.9	29.2	35.6	53.7	59.0	64.3	66.8	64.8	58.9	50.2	40.6	31.5
	5%	DB	25.1	27.3	36.6	63.5	78.1	83.7	84.3	81.2	73.4	57.6	39.7	29.6
		MCWB	23.8	25.8	33.9	50.2	57.7	63.6	66.4	63.4	56.4	47.5	36.8	28.2
	10%	DB	21.3	23.4	34.2	58.2	73.3	80.0	81.0	77.8	68.1	52.7	36.3	26.4
		MCWB	20.2	22.1	31.9	47.4	56.0	62.9	65.1	61.9	54.6	45.1	33.9	25.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.6	32.1	39.7	57.9	63.9	69.8	71.6	69.8	64.3	54.1	44.6	33.3
		MCDB	33.6	33.3	44.4	74.1	82.2	83.7	83.4	81.4	81.6	64.8	46.9	34.3
	2%	WB	28.1	29.3	36.3	54.5	61.2	67.3	69.5	66.9	60.2	51.5	40.4	31.6
		MCDB	29.2	30.8	39.3	67.1	77.6	80.0	81.5	79.1	75.0	60.9	43.8	32.8
	5%	WB	23.8	25.9	34.0	51.1	59.1	65.8	68.0	65.2	57.9	48.6	37.0	28.3
		MCDB	24.9	27.3	36.3	61.6	73.9	78.0	79.0	77.7	70.9	56.3	39.6	29.5
	10%	WB	20.2	22.2	32.2	48.1	57.1	64.3	66.7	63.4	55.5	45.6	34.1	24.9
		MCDB	21.3	23.4	34.2	57.1	71.5	76.5	77.2	75.0	66.5	52.2	36.2	26.3
Mean Daily Temperature Range	5% DB	MDBR	14.3	15.3	15.0	17.7	20.3	18.6	17.8	19.2	18.8	15.0	11.7	13.7
		MCDBR	15.0	14.2	13.1	25.4	26.4	24.2	22.4	25.1	28.3	23.9	12.3	14.1
		MCWBR	15.0	13.7	10.9	13.3	10.3	8.4	7.6	9.0	11.8	12.8	9.6	14.0
	5% WB	MCDBR	14.8	13.7	12.6	24.3	23.3	19.6	18.8	21.1	24.7	21.4	12.2	14.1
		MCWBR	14.9	13.3	10.8	13.8	10.7	8.3	7.6	8.8	11.6	12.6	9.9	14.1
Clear-Sky Solar Irradiance	taub	0.294	0.325	0.382	0.405	0.408	0.413	0.439	0.421	0.373	0.352	0.330	0.298	
	taud	2.029	1.979	1.926	2.159	2.208	2.278	2.207	2.226	2.310	2.253	2.101	1.991	
	Ebn at Noon	198	232	244	260	266	266	255	252	251	226	182	165	
	Edh at Noon	24	35	46	42	42	40	42	39	31	27	22	20	
All-Sky Solar Radiation	RadAvg	346	693	1175	1495	1843	1971	1779	1512	1081	566	272	228	
	RadStd	27	31	91	108	152	132	121	114	100	70	35	29	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page